PROJECT AND TEAM INFORMATION

Project Title

Localkart

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Born and brought up in a remote and hilly area, I personally witnessed that quick commerce platforms such as Blinkit and Bingo can deliver groceries and other essentials within hours. Zepto do not penetrate to places such as mine. My mother, who does not know how to drive a car, was struggling.My mother, who doesn't know how to drive a car, was struggling. He purchased all daily household needs for them as the local store was too far to walk to. This made me realize that, that those who do not own a vehicle (particularly in remote and hilly areas) do not have easy access to a vehicle. Shopping option – unlike those that live in the cities, who can shop and get the things at their doorsteps. LocalKart solves this by enabling local vendors to post and sell products on the web, while delivery agents carrying out orders. In their own local communities. Introducing speedy commerce convenience in places Blinkit and Zepto. While opening up local job opportunities for delivery persons, cannot reach. Additionally, LocalKart offers a pricing comparison feature unlike Blinkit, which offers fixed prices. and choose the one that's providing it at the lower price, making it convenient for the customers and saving them money as well. on their purchases.

State of the Art / Current solution

These days, apps such as Blinkit, Zepto, and Swiggy Instamart are bringing groceries and everyday essentials very close "to you". Rapidly in large urban areas only. The principle behind these applications is to maintain large warehouses near the customer. Where many people live locally, only makes sense to have a lot of customers, which is what has been done. Due to this, villages, the hill No service at all from them to areas and small towns. In these regions individuals still shop the old fashioned manner. — walking to the local shop, calling the shopkeeper, or sending a message on WhatsApp and hoping It can be delivered by somebody. No app to check out the products and compare prices, or track. the order. Meanwhile local traders miss out as well because they can only make sales to those who show up in person in their store, and have no means of reaching out to more customers or expanding their sales online, Unlike large retail stores that are now using apps to sell their products, the sellers here don't make any such use.

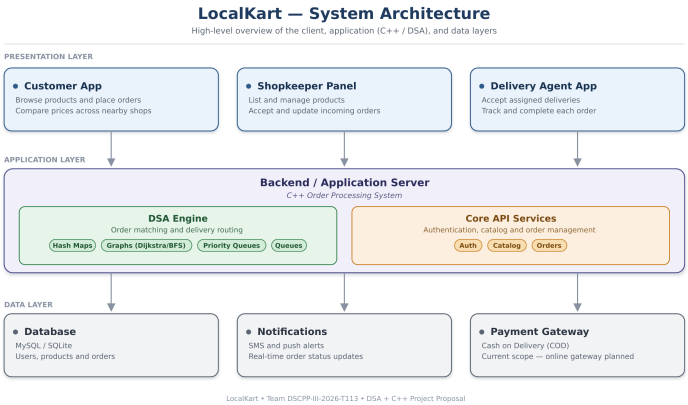
Project Goals and Milestones

LocalKart's primary use is to enable small, local shopkeepers to sell their products online and allow people to buy them. Eating healthy and ordering essentials like groceries, medicine, and cleaning supplies is easy for people in rural and remote communities, just as it is for people in cities by calling or using apps like. Blinkit. In many villages and areas of the hills, there is none at all, and the aim is to create a simple... A system that allows the shopkeeper to post his item and the customer to view the item and compare prices. Orders can be placed across shops and picked up and delivered in the local area by delivery agents. This way, shopkeepers have an opportunity to extend their business and customers get The same convenience which is already granted to the people in the cities. The aim of the project is:
1.Create a basic c++ system using basic dsa which will allow the user to open the shops,
2.Make a system to enable and allow the user to buy their products from shops and can compare prices from local shops and find closest delivery agent for every order with a a simple logic.
3.After the system is made and running , later we have to make a real time order tracking and monitoring system.So that the customer are aware of where their product is ,we have to make a rating system for shops and delivery agent and to build trust through it.
4.lastly we have to make a mobile app so that this project can actually be used by shopkeepers and customers.

Project Approach

The way I've been doing it, is to first develop the core logic, and then add the app. layer.
Step 1 – Core Logic (DSA + C++): Core Logic, will be designed using C++ and Data Structures. In just one example, I will use hash maps to have a quick product/shopkeeper lookup, and graphs to map delivery routes. The agents' priority is to be directed to nearby shops and customers (shortest path via Dijkstra/BFS for delivery routing), Queuing for order assignment based on distance/urgency and Linked lists/queues for order management flow: pending, accepted (pending on acceptance), delivered (pending on delivery).
Step 2 – Database: I will store the details of the shopkeeper, the list of products and the customer in a database, either MySQL or SQLite. Structured tables for orders, delivery agent information.
Now moving to Step 3 – Application Layer: I will create a simple front end (HTML/CSS/JS) or simple 'opening' where the customers can view the products, the shopkeepers can add the products to the list, and the delivery service can deliver. agent(s) is ready to receive orders.
In step 4 – Connecting everything, I will use a simple backend to connect the front-end to my c++ logic. Because it's combined with a C++ server or a light framework (like a C++ server or a lightweight framework), requests from the app will be processed by the actual DSA-based server. functions.
Step 5 – Testing: I'll test with some sample data – some fake shops, products and delivery routes – to verify that the order matching and delivery assignment process is correct and efficient.

System Architecture



Project Outcome / Deliverables

One of the primary results of this project is the creation of a working console based application called LocalKart which is able to link to local shop owners, customers and delivery staff all on a single platform. Unlike other apps such as Blinkit or Zepto, LocalKart is meant for smaller towns and regions that don't have such services. Gives shopkeepers an opportunity to sell online. This project's key deliverables are: A working C++ application for a retail store to add and control stock. A customer module, including the ability to browse items, place orders and track order status. A delivery agent module, for accepting and completing deliveries. Application of concepts of DSA including queues, linked lists, sorting for order and inventory. A report back regarding the system design, logic and the practical applicability of the proposed project. This is an overall project demonstrating application of DSA concepts to create a practical solution of real world. That will enable small retailers to attract a larger customer base.

Assumptions

- *Shopkeepers, customers, and delivery agents are assumed to be within the same local area or town.*
- *Only one delivery agent handles one order at a time.*
- *Product stock and prices are entered correctly by the shopkeeper.*
- *The system runs on a single system/console, without real internet or GPS tracking.*
- *Payment is assumed to be cash on delivery, so no online payment gateway is used.*

References

1. GeeksforGeeks, C++ Data Structures and Algorithms tutorials – https://www.geeksforgeeks.org/data-structures/
2. cppreference.com, C++ language and library documentation – https://en.cppreference.com/w/
3. Blinkit and Zepto official websites, referred to for understanding basic quick-commerce app workflow – https://www.blinkit.com
4. College/course DSA class notes and lab material provided during the semester.